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SCHOOL OF ENGINEERING AND TECHNOLOGY
CIVIL ENGINEERING DEPARTMENT

HYDRAULIC ENGINEERING OBSERVATION BOOK

BTech 4 Semester-2024-25

NAME OF THE STUDENT: ----------

Register No: ----------

School of Engineering & Technology-Kengeri Campus
Civil Engineering Department

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Ex.No.1	CALIBRATION OF V-NOTCH
Date:	

Aim: To determine co-efficient of discharge for a given triangular notch

Equipment and Instruments:

1. A channel fitted with vertical perforated plates to reduce the velocity of approach and arrangement for regulated supply of water.
2. Pointer gauge for measuring head of water over the crest.
3. A collecting tank fitted with a piezometer tube and graduated scale
4. ‘ V ‘ notch (Triangular Notch)
5. Stop watch.

Observations:

1. Area of Measuring tank A= 0.2 m^2
2. Apex angle of the V-notch, $\theta = 60^0$

Theory:

A notch may be defined as an operating provided in the side of a tank such that the surface in the tank is bellow the top edge of the operating. It is used to measure the flow rate of the fluid through a small channel or a tank. A weir is also same as notch but bigger in size and made up of concrete or masonry structure while notch is made up of metallic plate. The sheet of water flowing through a notch is known as nappe or weir. The bottom edge of the notch over which the water flow is still or crest and its height above the bottom of the tank or channel is known as crest height.

A triangular notch for measuring low discharge. Triangular notch is an ordinary notch which is having a triangular or V-shaped opening so that water is discharged through the opening only. Generally, a triangular notch is preferred to a rectangular notch for measuring low discharge because of its accuracy.

The discharge through a triangular notch is given by

$$Q_{theo} = \frac{8}{15} \times \tan\left(\frac{\theta}{2}\right) \times \sqrt{2g} \times H^{5/2} = \text{.....} \; m^3/\text{sec}$$

Procedure:

1. Note down the angle of the V-notch and dimensions of the collecting tank.
2. Open the valve and allow the water level to rise only up to sill of the notch.
3. Adjust the pointer gauge so that it coincides with the water surface and note the pointer gauge.
4. Open the inlet valve so that water flows over the notch to same height; wait for steady flow to take place.

5. Note this water level by means of pointer gauge.
6. The difference between h_1 and h_2 gives the head of water (H) over the notch.

TABLE OF READINGS

Sl/No:	Valve opening.	Head of water over crest, H, m				Collection of water discharged	
		Initial, mm	Final, mm	Difference in reading		Rise in water level in tank, R, mm	Time taken, t, s
				'h, mm	H, m		
1	¼ opening						
2	½ opening						
3	¾ opening						
4	Full opening						

Specimen Calculations:

Actual Discharge $Q_{act} = \frac{A \; x \; R}{t \; x \; 1000}$ m^3/s

Theoretical Discharge $Q_{theo} = \frac{8}{15} \times \tan\left(\frac{\theta}{2}\right) \times \sqrt{2g} \times H^{5/2} = \ldots\ldots\ldots\ldots m^3/\text{sec}$

Co-efficient of discharge $C_d = \frac{Q_{act}}{Q_{theo}} = \ldots\ldots\ldots\ldots$

TABLE OF CALCULATION

Sl/No:	Valve opening	Actual Discharge, m^3/s	Theoretical Discharge, m^3/s	Coefficient of discharge, C_d.
1	¼ opening			
2	½ opening			
3	¾ opening			
4	Full opening			

RESULT:

The coefficient of discharge of V- notch has is determined at different discharge levels.

ROUGH WORK

Ex.No.2 Date:	CALIBRATION OF RECTANGULAR NOTCH

Aim: To determine co-efficient of discharge for a given Rectangular notch

Equipment and Instruments:

A channel fitted with vertical perforated plates to reduce the velocity of approach and arrangement for regulated supply of water., Pointer gauge for measuring head of water over the crest, A collecting tank fitted with a piezometer tube and graduated scale and Rectangular Notch Plate

Observations:

1. Area of Measuring tank A= 0.2 m^2
2. Width of the Notch normal to flow, b =

Theory:

A notch may be defined as an operating provided in the side of a tank such that the surface in the tank is below the top edge of the operating. It is used to measure the flow rate of the fluid through a small channel or a tank. A weir is also same as notch but bigger in size and made up of concrete or masonry structure while notch is made up of metallic plate. The sheet of water flowing through a notch is known as nappe or weir. The bottom edge of the notch over which the water flow is still or crest and its height above the bottom of the tank or channel is known as crest height.

A rectangular notch is used relatively large discharges. Rectangular notch is an ordinary notch which is having a rectangular opening so that water is discharged through the opening only. Shape of the nappe changes with respect to the head over rectangular notch.

The discharge through a rectangular notch is given by

$$\boldsymbol{Q_{theo} = \frac{2}{3} \times b \times \sqrt{2g} \times H^{\frac{3}{2}} = \ldots\ldots\ldots\ldots \frac{m^3}{sec}}$$

Procedure:

1. Note down the width of the rectangular-notch and dimensions of the collecting tank.
2. Open the valve and allow the water level to rise only up to sill of the notch.
3. Adjust the pointer gauge so that it coincides with the water surface and note the pointer gauge.
4. Open the inlet valve so that water flows over the notch to same height; wait for steady flow to take place.
5. Note this water level by means of pointer gauge.

6. The difference between h_1 and h_2 gives the head of water (H) over the notch.

TABLE OF READINGS

Sl/No:	Valve opening.	Head of water over crest, H, m				Collection of water discharged	
		Initial, mm	Final, mm	Difference in reading		Rise in water level in tank, R, mm	Time taken, t, s
				‘h, mm	H, m		
1	¼ opening						
2	½ opening						
3	¾ opening						
4	Full opening						

Specimen Calculations:

Actual Discharge $Q_{act} = \frac{A\ x\ R}{t\ x\ 1000}$ m^3/s

Theoretical Discharge $\boldsymbol{Q_{theo} = \frac{2}{3} \times b\sqrt{2g} \times H^{\frac{3}{2}} = \ldots\ldots\ldots\ldots \frac{m^3}{sec}}$

Co-efficient of discharge $C_d = \frac{Q_{act}}{Q_{theo}} = \ldots\ldots\ldots\ldots$

TABLE OF CALCULATIONS

Sl/No:	Valve opening	Actual Discharge, m^3/s	Theoretical Discharge, m^3/s	Coefficient of discharge, C_d.
1	¼ opening			
2	½ opening			
3	¾ opening			
4	Full opening			

RESULT:

The coefficient of discharge of Rectangular- notch has been determined at different discharge levels.

ROUGH WORK

Ex.No.3 & 4	**CALIBRATION OF BROAD CRESTED and OGEE WEIR**
Date:	

Objective: To determine the coefficient of discharge of Ogee and Broad crested weir.

Theory: A weir is a notch on a large scale used for measuring the flow of a river, canal etc

Broad Crested Weir: A weir is said to be broad crested weir, if the streamlines are parallel to the surface of the crest. The width of the crest of the broad crested weir is in between 2.5H and 10H.

Ogee Weir: A weir is said to be a Ogee weir, if the downstream surface has ogee profile. It is used in spillway. The main advantage of providing such a shape for the spillway is that the flowing sheet of water remains in contact with the surface of the spillway, thereby preventing the negative pressure being developed on the downstream side. However, this condition will be fulfilled only as long as the head over the crest of the spillway is equal to or less than the designed head.

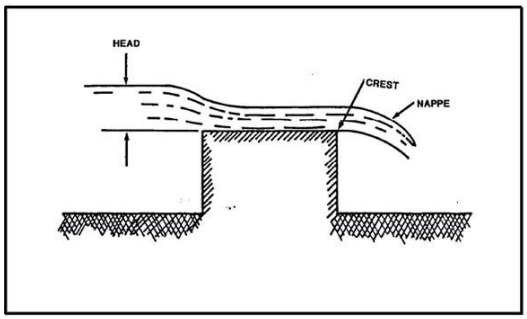


Fig. Broad crested weir

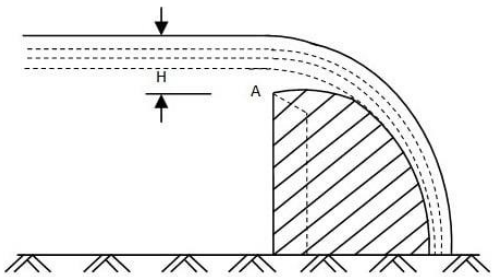


Fig. Ogee weir

Procedure:

1. Note down the length of the broad crested weir/Ogee weir (L) and dimensions of the collecting tank (b and d).
2. Switch on the pump and open the valve and allow the water level to rise only up to sill of the weir.
3. Note down the sill level with the help of the point gauge
4. Open the inlet valve so that water flows over the weir to some height; wait for steady flow to take place and note down the head over the weir (hf).
5. Note down the time (t) taken for R cm rise of water in the collecting tank to determine the actual discharge.
6. Repeat step 4 and 5 for various discharges and tabulate the obeservations
   .

The difference between hi and hf gives the head of water (H) over the weir

Formulae involved: Broad Crested Weir

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Actual Discharge $Q_{act} = \frac{A \, x \, R}{t \, x \, 1000}$ m^3/s

Theoretical Discharge $Q_{th} = 1.705 \times L \times H^{\frac{3}{2}} =$ ..... $\frac{m^3}{sec}$

Co-efficient of discharge $C_d = \frac{Q_{act}}{Q_{th}} =$.....

Observations:

Width of the broad crested weir (b)= ----------m

Length of the broad crested weir (L) = ---------- m

Height of Broad crested weir = ---------- m

Length of Ogee weir = ---------- m

Area of the Collecting Tank= ---------- m^2

Table of Observations- Broad Crested Weir

Sl/No:	Valve opening.	Head of water over crest, H, m				Collection of water discharged	
		Initial, mm	Final, mm	Difference in reading		Rise in water level in tank, R, mm	Time taken, t, s
				h (mm)	H (m)		
1	¼ opening						
2	½ opening						
3	¾ opening						
4	Full opening						

Table of calculations: Broad crested Weir

Sl/No:	Valve opening	Actual Discharge, m^3/s	Theoretical Discharge, m^3/s	Coefficient of discharge, C_d.
1	¼ opening			
2	½ opening			
3	¾ opening			
4	Full opening			

Sample calculations: Broad Crested Weir

H=

Qth =

Qact =

Cd=

Formulae involved: Ogee weir

Actual Discharge $Q_{act} = \frac{A \; x \; R}{t \; x \; 1000}$ m^3/s

Theoretical Discharge $Q_{theo} = \frac{2}{3} \times L\sqrt{2g} \times H^{\frac{3}{2}} =$ ..... $\frac{m^3}{sec}$

Co-efficient of discharge $C_d = \frac{Q_{act}}{Q_{theo}} =$ .....

Table of Observations- Ogee Weir

Sl/No:	Valve opening.	Head of water over crest, H, m				Collection of water discharged	
		Initial, mm	Final, mm	Difference in reading		Rise in water level in tank, R, mm	Time taken, t, s
				h (mm)	H (m)		
1	¼ opening						
2	½ opening						
3	¾ opening						
4	Full opening						

Table of calculations: Ogee Weir

Sl/No:	Valve opening	Actual Discharge, m^3/s	Theoretical Discharge, m^3/s	Coefficient of discharge, C_d.
1	¼ opening			
2	½ opening			
3	¾ opening			
4	Full opening			

Sample calculations: Ogee Weir

H=

Qth =

Qact =

Cd=

Ex.No.5	CALIBRATION OF ORIFICEMETER
Date:	

Aim: To determine the co-efficient of discharge (C_d) for given Orificemeter.

Equipment and Instruments:

1. Regulated supply of water arrangement.
2. A Orificemeter connected to a U tube.
3. A measuring tank fitted with a piezometer tube and a graduated scale.
4. Stop watch.

Observations:

1. Area of Measuring tank, A= 0.12 m^2
2. Diameter of the Orifice meter, d= 12.5 mm
3. Diameter of the inlet pipe of the Orifice meter, D= 25 mm
4. Acceleration due to Gravity, g= 9.81 m/sec^2

Theory:

It is a device used for measuring the rate of flow of a liquid through a pipe. I t is a cheaper device as compared to venture meter. This is a practical example for Bernoulli's equation. It consists of a flat circular plate which is concentric with pipe. The orifice die is kept generally 5 times the die of the pipe through it may vary from 4 to 8 times the pipe die.

Two pressure tapings are provided at section (1) which is located at a distance of 1.5 times to 2 times the pipe dia upstream from the orifice plate and other at section (2) which is located at a distance of half die of the orifice on the downstream side of the orifice plate. Since the dia of the orifice is less than the pipe dia as the fluid flow through the orifice the flowing stream converges which results in acceleration of the flowing fluid in accordance with the consideration of continuity. The effect of the converges of the flowing fluid stream extends up to a certain distance upstream from the orifice plate and therefore the pressure and taping on the upstream side is provided away from the orifice plate of a section where this effect is nonexistent. However, on the downstream side the pressure tapings are provided quite close to the orifice plate at the section where the converging jet of the fluid has almost the cross-section area and resulting in almost max. Velocity of flow and consequently the min pressure at the sections.

Discharge through an orifice meter is given by

$$Q_{the} = \frac{a_1 a_2 \sqrt{2gH}}{\sqrt{(a_1^2 - a_2^2)}} = \ldots\ldots\ldots\ldots\ldots\ldots m^3/\text{sec}$$

Where $a_1 = \frac{\pi D^2}{4}$ = Area of inlet Section of Orificemeter m^2, $a_2 = \frac{\pi d^2}{4}$ = Area of Orificemeter (Throat) m^2

Procedure:

1. Select the required Orifice meter and note down the dimension at inlet and outlet.

2. Open the flow contact valve and allow a certain discharge.
3. Remove the air bubbles from the manometer limbs.
4. Note down the measuring levels on the two limbs of a manometer.
5. Note down the diameter of the collecting tank and collect the water in it by closing the drain valve and time taken for a known rise of water in tank say 10 cm.
6. Repeat the step 3, 4 and 5 for different trial

Specimen Calculations:

1. Theoretical Discharge $Q_{the} = \frac{a_1 a_2 \sqrt{2gH}}{\sqrt{(a_1^2 - a_2^2)}} = \ldots\ldots\ldots\ldots\ldots\ldots m^3/\text{sec}$

Where $a_1 = \frac{\pi D^2}{4}$ = Area of inlet Section of Orifice meter m^2

$a_2 = \frac{\pi d^2}{4}$ = Area of Orifice meter m^2

$H = 12.6\ x\ \frac{h}{1000}$ m

2. Actual Discharge

$Q_{act} = \frac{A\ x\ R}{t\ x\ 1000}$ m^3/s

Where A= Area of measuring tank in m^2

R= Rise of water level in meters

t= Time taken in seconds

3. Co-efficient of Discharge C_d

$$C_d = \frac{Q_{act}}{Q_{the}} = \frac{Actual Disch\arg e}{Theoretical Dish\arg e} = ...$$

TABLE OF READINGS

Valve opening	**Rise in water level in collection tank, R, mm**	**Time taken, t, s**	**Differential Head, mm of Hg**
¼ opening			
½ opening			
¾ opening			
Full opening			

TABLE OF CALCULATION

Valve opening	**Theoretical Discharge, Q_{th}, m^3/s**	**Actual Discharge, Q_{act}, m^3/s**	**Coefficient of discharge, C_d**
¼ opening			
½ opening			
¾ opening			
Full opening			

Result:

The coefficient of discharge of an Orifice Meter has been determined.

Rough work

Ex.No.6 Date:	CALIBRATION OF VENTURI METER

AIM :

To demonstrate the use of Venturimeter as flow meter and to determine the co-efficient of discharge.

APPARATUS:

1. Measuring Tank to measure the flow rate.
2. A pipe line with a Venturimeter.
3. Tappings with Ball Valves are provided at Inlet & Throat of Venturimeter connected to Manometer.
4. A Constant steady supply of water with a means of varying the flow rate using Monobloc pump.
5. Valve is provided for Venturimeter to conduct experiments.
6. Electronic Digital Timer with Float Switch for measurement of Flow Rate by collecting fixed quantity of water. (This arrangement is optional, if provided).

THEORY:

A *VENTURI METER* is a device which is used for measuring the rate of flow of fluid through pipe line. The basic principle on which a venturimeter works is that by reducing the cross – Sectional area of the flow passage, a pressure difference is created between the inlet and throat & the measurement of the pressure difference enables the determination of the discharge through the pipe.

CONSISTS OF :

A Venturimeter consists of ,

1. An Inlet Section followed by a Convergent Cone,
2. A Cylindrical Throat &
3. A gradually Divergent Cone.

The Inlet section of the Venturimeter is of the same diameter as that of the pipe which is followed by a Convergent Cone. The Convergent Cone is a short pipe which tapers from the original size of the pipe to that of the throat of the Venturimeter. The throat of the Venturimeter is a short parallel side tube having its cross sectional area smaller than that of the pipe. The Divergent Cone of the Venturimeter is gradually diverging pipe with its cross sectional area increasing from that of the throat to the original size of the pipe. At the inlet section & the throat, of the venturimeter, pressure taps are provided through pressure rings.

FORMULAE

DATA :

*	Area of Measuring Tank,	" A "	=	0.12 m^2
*	Acceleration due to gravity,	" g "	=	9.81 m/sec^2
*	Diameter of the Venturimeter (Throat),	" d "	=	12.5 mm.
*	Diameter of the Inlet pipe of the Venturimeter	" D "	=	25 mm.

1. Theoretical Discharge $$Q_{the} = \frac{a_1 a_2 \sqrt{2gH}}{\sqrt{(a_1^2 - a_2^2)}} = \text{.....} \; m^3/\text{sec}$$

Where $$a_1 = \frac{\pi D^2}{4} = \text{Area of inlet Section of venturimeter} \quad m^2$$

$$a_2 = \frac{\pi d^2}{4} = \text{Area of venturimeter} \quad m^2$$

$$H = 12.6 \; x \; \frac{h}{1000} \qquad m$$

4. Actual Discharge

$$Q_{act} = \frac{A \; x \; R}{t \; x \; 1000} \qquad m^3/s$$

Where A= Area of measuring tank in m^2

R= Rise of water level in meters

t= Time taken in seconds

5. **Co-efficient of Discharge C_d**

$$C_d = \frac{Q_{act}}{Q_{the}} = \frac{Actual Disch\arg e}{Theoretical Dish\arg e} = ...$$

TABLE OF READINGS

Valve opening	**Rise in water level in collection tank, R, mm**	**Time taken, t, s**	**Differential Head, mm of Hg**
¼ opening			
½ opening			
¾ opening			
Full opening			

TABLE OF CALCULATION

Valve opening	**Theoretical Discharge, Q_{th}, m^3/s**	**Actual Discharge, Q_{act}, m^3/s**	**Coefficient of discharge, C_d**
¼ opening			
½ opening			
¾ opening			
Full opening			

Result:

The coefficient of discharge of a Venturimeter has been determined.

Rough Work

Rough Work

Ex.No.7	DETERMINATION OF DARCY'S FRICTION FACTOR FOR A STRAIGHT PIPE.
Date:	

AIM: To determine the Head Loss due to Friction in Pipe Length and hence, to determine the "Friction in Pipe".

APPARATUS:

1. The Apparatus consists of 4 different Pipe lines of 1" GI Pipe, 1" S.S Pipe, 3/4" GI Pipe and 1/2" GI Pipe.
2. U- Tube Manometer connected across each pipeline with change –over valves.
3. A constant steady supply of water using Centrifugal Pump with Valve for varying the flow rate.
4. Measuring Tank with Stop Clock for measurement of flow rate.
5. The Sump Tank mounted below the Collecting Tank.
6. The Butterfly Valve for instantaneous dumping of collected water into the sump tank, and the overflow pipe to facilitate to run the equipment on closed circuit basis.

THEORY:

The straight pipes produce the friction to the flow of fluid due to its inside roughness and also offer Resistance /Friction to the flow of fluid. The head loss due to friction in straight pipes is expressed by the standard formulae:

$$h_f = \frac{4fLV^2}{2gd} \qquad \text{..... (a)}$$

SPECIFICATIONS:

- Pump Capacity : 1 HP, 1Ph
- Collecting (Measuring) Tank Area : 0.12 m^2
- Nominal Dia of Pipes : 1 NB (26mm ID)
  : 1" GI Pipe
  : 1" S.S Pipe
  : 3/4" GI Pipe
  : 1/2" GI Pipe

OPERATING PROCEDURES:

All the necessary instrumentation along with its accessories are readily connected. It is just enough to follow the instructions below:

1. Fill- in the sump tank with clean water.
2. Keep the delivery valve closed.
3. Connect the power cable to 1 Ph, 220 V, 10 Amps with earth connection.
4. Switch – ON the Pump & open the delivery valve.
5. Open the corresponding ball valve of the pipe line, to be tested.
6. Adjust the flow through the control valve of the pump.

7. Open the corresponding ball valves.
8. Note down the differential head reading in the Manometer. (Expel if any air is there by opening the air vent valve provided with the Manometer.)
9. Operate the Butterfly valve of the collecting tank for rise of water level and note down the time and keep it open when the readings are not taken.
10. Change the flow rate & repeat the experiment for different diameter of pipes.

PRECAUTIONS AND THINGS TO REMEMBER:

1. Do not start the pump if the voltage is less than 180V.
2. Do not forget to give electrical neutral & earth connections correctly.
3. Frequently (at least once in three months) Grease / oil the rotating parts.
4. Initially put clean water free from foreign material, and change the water once in three months.
5. At least once every week, operate the unit for five minutes to prevent clogging of the moving parts.

FORMULAE

1. Actual Discharge $$Q = \frac{A \, x \, R}{1000 \, x \, t} \quad m^3 / Sec.$$

Where A= Area of collecting tank = 0.12 m^2
R= Rise of water in collecting tank in mm
t = Time taken in seconds for R mm water collected

2. Loss of Head due to Friction $$h_f = \frac{(13.6 - 10) x H}{1000} \quad m$$

Where 'H' = Difference in Mercury Column in **mm** of Hg in Double Column Manometer.

3. Velocity

$$V = \frac{Discharge}{Area\ of\ cross-section\ of\ pipe} = \frac{Q}{a} \quad m/S$$

Where 'a' = Area of the pipe $= \frac{\pi}{4} d^2$ m^2

'd' = Inner Dia of the Pipe. in meter

4.

Friction Factor $$f = \frac{h_f \, g d}{2LV^2}$$

TABLE OF READINGS

Type of pipe		DISCHARGE OF WATER COLLECTED		DIFFERENTIAL HEAD 'H' in mm of Hg.
		Rise in Water Level 'R' in mm	Time Taken in Secs. 't'	
Diameter 16 mm (½" pipe)	Quarter opening of valve	100		
	Half opening of valve	100		
	Full opening of valve	100		
Diameter 21 mm (¾" pipe)	Quarter opening of valve	100		
	Half opening of valve	100		
	Full opening of valve	100		
Diameter 27 mm (1" pipe)	Quarter opening of valve	100		
	Half opening of valve	100		
	Full opening of valve	100		

TABLE OF CALCULATIONS

Type of Pipe		Loss of Head h_f	Velocity m /Sec.	Friction factor 'f'
Diameter 16 mm (½" pipe)	Quarter opening of valve			
	Half opening of valve			
	Full opening of valve			
Diameter 21 mm (¾" pipe)	Quarter opening of valve			
	Half opening of valve			
	Full opening of valve			
Diameter 27 mm (1" pipe)	Quarter opening of valve			
	Half opening of valve			
	Full opening of valve			

Result:

The head loss in pipes of different diameters due to friction (Major Losses) has been determined.

Rough work

Ex.No.8 Date:	DETERMINATION OF MINOR LOSS CONSTANTS

AIM: To determine the minor losses in a pipe.

INTRODUCTION:

When a fluid flow through a pipe, certain resistance is offered to the flowing fluid, which results in causing a loss of energy. The various energy losses in pipes may be classified as ;

(i) Major Losses.
(ii) Minor Losses

The major losses of energy, as a fluid flow through a pipe, are caused by friction. It may be computed by Darcy- Weisbach Equation as indicated earlier. The loss of energy due to friction is classified as a Major Loss because in case of long pipelines it is usually much more than the loss of energy incurred by other causes.

The Minor Losses of energy are those which are caused on account of the change in the velocity of flowing fluid (either in magnitude or direction). In case of long pipes these losses are usually quite small as compared with the loss of energy due to friction and hence these are termed 'Minor Losses' which may even be neglected without serious error. However, in short pipes these losses may sometimes out weight the friction loss. Some of the losses of energy which may be caused due to the change of velocity are indicated below:

a) Loss of Energy due to sudden Enlargement,

$$h_L = \frac{(V_1 - V_2)^2}{2g}$$

b) Loss of Energy due to sudden contraction,

$$h_L = 0.5\,\frac{V_2^2}{2g}$$

c) Loss of Energy in Bends,

$$h_L = 0.5\,\frac{V_2^2}{2g}$$

d) Loss of Energy in Various pipe Fittings (Elbows, Valves, Collars)

$$h_L = 0.6\,\frac{V_2^2}{2g}$$

SPECIFICATIONS:

* Pump Capacity : 1HP, 1 ph.

* Collecting (Measuring) Tank Area : 0.12 m^2

* Pipe Fittings : Contraction, Expansion, 90^0 Long Bend & 90^0 elbow

OPERATING PROCEDURES:

All the necessary instrumentation along with its accessories are readily connected. It is just enough to follow the instructions below:

1. Fill in the sump Tank with clean water.
2. Keep the delivery valve closed.
3. Connect the power cable to 1 ph, 220V, 10Amps with Earth Connection.
4. Switch ON the pump & open the Delivery Valve.
5. Open the corresponding Ball Valve of the Pipe fittings.
6. Adjust the flow through the Control Valve of the pump.
7. Open the corresponding Ball Valves.
8. Note down the differential head reading in the Manometer. (Expel if any air there by opening the Drain Cocks provided with the Manometer.)
9. Operate the Butterfly Valve to note down the collecting tank reading against the known time and keep it open when the readings are not taken.
10. Change the flow rate & repeat the experiment for different pipe Fittings.

FORMULE FOR PIPE FITTINGS

FORMULAE

1. Actual Discharge $$Q = \frac{A \; x \; R}{1000 \; x \; t} \quad m^3 / Sec.$$

Where A= Area of collecting tank = 0.12 m^2
R= Rise of water in collecting tank in mm
t = Time taken in seconds for R mm water collected

2. Loss of Head due to Fitting, $$h_f = \frac{(13.6 - 1.0)xH}{1000} \quad m$$

Where 'H' = Difference in Mercury Column in **mm** of Hg in Double Column Manometer.

3. Velocity

$$V = \frac{Discharge}{Area\ of\ cross-section\ of\ pipe} = \frac{Q}{a} \qquad m/S$$

Where 'a' = Area of the pipe $= \frac{\pi}{4} d^2$ m^2

'd' = Inner Dia of the Pipe. in meter

a) Loss of Energy due to sudden Enlargement,

$$h_L = \frac{(V_1 - V_2)^2}{2g}$$

b) Loss of Energy due to sudden contraction,

$$h_L = 0.5 \frac{V_2^2}{2g}$$

c) Loss of Energy in Bends,

$$h_L = 0.5 \frac{V_2^2}{2g}$$

Pipe Sizes (inner diameters):

1. **Sudden expansion:**
   a. **Inlet dia = 16 mm**
   b. **Outlet dia = 27 mm**
2. **Sudden contraction:**
   a. **Inlet dia = 27 mm**
   b. **Outlet dia = 16 mm**
3. **Bend:**
   a. **Pipe dia = 27 mm.**

TABLE OF READINGS

TYPE OF FITTING		DISCHARGE OF WATER COLLECTED		DIFFERENTIAL HEAD 'H' (Manometer Reading)in mm of H_q.
		Rise in water Level 'R' in mm	Time Taken "t" in Secs.	
Sudden contraction	Quarter opening of valve	100		
	Half opening of valve	100		
	Full opening of valve	100		
Sudden expansion	Quarter opening of valve	100		
	Half opening of valve	100		
	Full opening of valve	100		
Bend	Quarter opening of valve	100		
	Half opening of valve	100		
	Full opening of valve	100		

TABLE OF CALCULATIONS

Type of Pipe		Loss of Head h_f	Velocity m /Sec.		Loss of Energy, H_L, m
			V_1	V_2	
Sudden expansion	Quarter opening of valve				
	Half opening of valve				
	Full opening of valve				
Sudden contraction	Quarter opening of valve				
	Half opening of valve				
	Full opening of valve				
Bend	Quarter opening of valve				
	Half opening of valve				
	Full opening of valve				

Result:

The head loss in pipes due to different fittings (minor losses) has been determined.

ROUGH WORK

Ex.No.9	DETERMINATION OF VANE COEFFICIENT FOR FLAT AND HEMISPHERICAL VANES.
Date:	

Aim: To determine the force exerted by a jet of water on a stationery Vane, and to verify the impulse momentum equation.

Description of apparatus:

1. The apparatus consists of a high efficient nozzle fitted to a 25 mm diameter supply with a gate valve.
2. The valves are fitted to the vertical bracket of the differential lever which is to balance the upward force of the jet.
3. The lever is balanced by metric weights. These are placed on the hanger of the lever. The unit is mounted on a strong Iron stand.
4. The discharged water can be collected in a measuring tank.

Observations:

1. Area of collecting tank A= 0.09 m^2
2. Diameter of the nozzle d= 4 mm
4. Hemispherical Vane angle, $\theta = 138^0$
5. Angle of inclined vane, $\theta = 60^o$.

6. Weight of vane assembly = 200 g

Theory:

The liquid comes out in the form of a jet from the outlet of a nozzle which is fitted to a pipe the liquid is flowing under pressure. Impact of jet means the force exerted by the jet on a plane which may be stationary or moving.

Let the jet strikes the smooth symmetrical stationary curved vane at the centre on the concave side. The jet after striking the plate comes out with the same velocity if the plate is smooth and there is no loss of energy due to impact of jet in the tangential direction of the curved vane. Let 'v' be the velocity of the jet after striking the plate θ be the angle between two tangents drown to the vanes at its outlet line. Thus the jet after striking will be deflected on each through on angle (180-θ) the velocity of the plate can be resolved into two components one in the direction of jet and other perpendicular to the direction of the jet. Component of velocity in the direction of jet = - v cosθ. Component of velocity in the direction perpendicular to the direction of jet is given by v sinθ.

For this experimental only, the force by the jet are needed and are needed and are

1. Flat vane,θ = 90°, $F = \rho \times a \times V^2$
2. Hemispherical vane,θ = 0, $F = 2 \times \rho \times a \times V^2$
3. Inclined vane, θ = 60°, F = ρ x a x V^2 sinθ

Procedure:

1. Fix any one type of vane.
2. Ensure that the lever is in horizontal plane.
3. Plane some weight on the hanger plate. Due to this the lever will get deflected.
4. Open the supply valve and gradually adjust the flow such hat the lever is brought back to flow horizontal position.
5. Note down the time required for known rise of water level in the measuring tank.
6. Repeat above procedure for different weight and then by changing the vanes also.

TABLE OF READINGS

Vane Type	Sl/No:	Rotameter Reading in lpm, (Q)	Force on the balance, F_a, g
Flat Vane	1		
	2		
	3		
Hemispherical vane	4		
	5		
	6		
Inclined vane	7		
	8		
	9		

Specimen Calculations:

1. **Actual Discharge** $Q_{act} = Q\ x\ \frac{10^{-3}}{60}$ m^3/s
2. **Area cross section of the nozzle,**

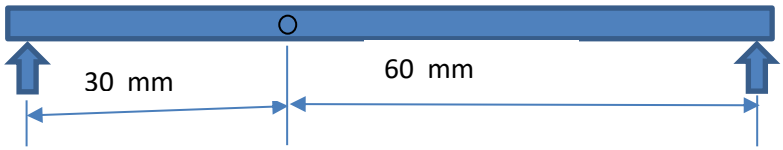
$$a = \frac{\pi}{4} x \left(\frac{4}{1000}\right)^2 = 1.2566\ x\ 10^{-5} \quad m^2$$

3. **Velocity of water in jet,** $V = \frac{Q_{act}}{a}$ m/s
4. **Actual force felt on the vane,** $F_{act} = (F_a\ x\ 2 + 200)\ x\ \frac{9.81}{1000}$ N

F_a is multiplied with 2 to account for the lever arm effect of the load transfer lever at the top of the balance.

Load transfer lever

End of lever at which the vane force is acting.



End of lever that is connected to the balance.

5. Theoretical force exerted by a jet of water on a Flat Vane

$$F_{theor} = \rho \; x \; a \; x \; V^2 \qquad N$$

6. Force exerted by a jet of water on a Hemispherical Vane

$$F_{theor} = 2 \; x \; \rho \; x \; a \; x \; V^2 \qquad N$$

7. Force exerted by a jet of water on an inclined vane

$$F_{theor} = \rho \; x \; a \; x \; V^2 \; x \sin 60$$

8. Coefficient of Impact, $CI = \dfrac{F_{act}}{F_{theor}}$

TABLE OF CACLULATION

Type of vane	Sl/No:	Actual Discharge, Q_{act}, m^3/s	Impact Velocity, V, m/s	Actual force, F_{act}, N	Theoretical force, F_{theor}, N	Coefficient of Impact, CI
Flat Vane	1					
	2					
	3					
Hemispherical vane	4					
	5					
	6					
Inclined vane	7					
	8					
	9					

Comment: The Coefficient of impact of for inclined vane is found to be more than 1. The value is more than 1 because the vane is a double inclined vane in which the vertical velocity of water is brought to zero value at the joint between the two inclined planes of the vane.

Result

The coefficient of impact for flat, hemispherical and inclined vanes have been determined.

Rough work

Ex.No.10	DETERMINATION OF HYDRAULIC COEFFICIENT OF A VERTICAL ORIFICE
Date:	

Objective: To determine the Coefficient of velocity, Coefficient of discharge, and coefficient of contraction.

Experimental Setup: orifice plate fitted to water tank, piezometer to determine the constant head, Collecting tank fitted with Piezometer to determine the actual discharge, scale to measure the coordinates of jet from vena-contracta and Stop watch.

Procedure:

1. Note down the dimensions of the collecting tank, diameter of the orifice.
2. Switch on the main supply switch of the pump connected to orifice setup
3. Allow some water to flow and maintain constant head (H) in the overhead tank
4. Measure Horizontal distance (x) and vertical distance (y) of the jet from the vena-contracta using horizontal and movable vertical scale under laminar flow conditions
5. Allow the water to flow into the collecting tank and note down the time taken (t sec) for R cm rise.
6. Repeat the same procedure 3 to 5 for variable discharges.

Calculations:

a= c/s are of orifice plate

H= Constant head measured from the center of orifice

A = c/s area of collecting tank

R= Rise of water R cm (10 cm)

t= time take for R cm rise in collecting tank

x = horizontal distance travelled by the jet in time t' = v *t' ----------(1)

y= vertical distance of jet = 0.5 gt'^{2} ---------- (2)

From (1) t' = x/v ----------(3)

Substitute t' in (2)

$$y = \frac{1}{2} g \frac{x^2}{v^2}$$

$$v = \sqrt{\frac{gx^2}{2y}} \text{ ----------(4)}$$

Where v = =v_a = actual velocity of jet

We know $v_{th} = \sqrt{2gH}$ ----------(5)

From (4) and (5) determine Coefficient of velocity (C_v)

$$C_v = \frac{v_a}{v_t} = \frac{\sqrt{\frac{gx^2}{2y}}}{\sqrt{2gH}} = \frac{x}{\sqrt{4yH}} \quad \text{---------- (6)}$$

$$C_d = \frac{Q_{act}}{Q_{th}} = \frac{\frac{AR}{t}}{a\sqrt{2gH}} \quad \text{----------(7)}$$

We know $C_d = C_v \, x \, C_c$

$$C_c = \frac{C_d}{C_v} \quad \text{----------(8)}$$

Table of Calculations

S.NO	Constant Head (H)	x	y	Time taken for Rcm rise in collecting tank (t)	$C_v = \frac{x}{\sqrt{4yH}}$	Qact $=\frac{AR}{t}$	Qth $=a\sqrt{2gH}$	$C_d = \frac{Q_{act}}{Q_{th}}$	$C_c = \frac{C_d}{C_v}$

Sample Calculations

Coefficient of Discharge (Cd) =

Coefficient of Velocity (Cv)=

Coefficient of Contraction (Cc)=

Result:

ROUGH WORK

Ex.No.11 Date:	PERFORMANCE TESTS ON A SINGLE STAGE OR MULTISTAGE CENTRIFUGAL PUMP

Aim: To obtain main and operating characteristics of a centrifugal pump.

Centrifugal Pump Specifications:

- Electrical supply : 230V, 15A, AC, 1 Phase, 50 Hz with Neutral & Earth Connections.
- Motor : AC Motor, 1.5HP, 1500RPM
- Pump : Kirloskar Make Centrifugal Pump. Max. Speed : 2880 RPM.
- Pressure Gauges : 2 Kg / Cm^2
- Vacuum Guage : 0-760 mm of Hg.
- Energy Meter Constant : 1200 Rev. / KWH.
- Speed indicator : 0-9999 RPM (Digital Type).
- Control Valve : For Suction & Delivery
- Measuring (Metering) Tank base area : 0.25m^2

Operating Procedure:

It is possible to run the pump at different speeds by shifting the V-belt location in the pulleys. There are three groves on the pulleys, hence, pump can be operated at three different speeds. After changing the belt location, belt tension is adjusted using the belt tension adjusting mechanism.

A. To Obtain Main Characteristics

This test is carried out by running the pump at different speeds and different discharges. Different speeds are achieved by shifting the V-belt location on the pulley. Different discharges are achieved by adjusting the delivery valve.

Step – 1: Keep the centrifugal pump delivery gate valve closed.

Step – 2: Switch on the centrifugal pump.

Step – 3: Open the delivery valve slightly so that the delivery pressure is readable.

Step – 4: Operate the butterfly valve of the collection tank to find out the time for 100 mm rise of water level in the collection tank. Note down the rpm of the pump.

Step – 5: Note down the reading of delivery side pressure gauges, suction side vacuum gauges, and time for 20 revolutions of energy meter disc.

Step – 6: Open the delivery valve further so that discharge is increased. Repeat Step – 4 to 5.

Step – 7: Repeat above steps for 5 discharge (minimum) rates for a particular rpm of the pump.

Step – 8. Change the rpm of the pump by shifting the V-belt to the adjacent grove.

Step – 9: Repeat Step – 4 to Step – 8 till the pump is run at all possible speeds.

Tabulate the readings as shown in Tabular Column – 1.

B. To Obtain Operating Characteristics

This test is carried out a by running the pump at a constant speed and at different discharges.

A separate test is not needed for obtaining Operating Characteristics. The reading taken for the test to obtain Main Characteristics at the pump speed of 2475 rpm is to be used.

Plot the graphs as shown below. Discharge is marked in X- axis and in the Y – axis, efficiency, pump input power, pump output power and delivery head are marked.

Calculation:

1. Total Head 'H' in m

$$H = 10 \text{ (Delivery Pressure + Vacuum Head)}$$

$$= 10 (H_d + H_v / 760).$$

Where,' H_d ' is the pressure in Kg/cm^2 read from Column (5)

' H_v ' is the Vacuum in mm of Hg. read from Column (4)

2. Discharge Rate 'Q' in m^3 / Sec.

$$Q = \frac{A \ x \ h}{1000 \, x \, t_w} = \frac{0.25 \ x \ 100}{1000 \, x \, t_w}$$

Where, 'A' = 0.25 m^2 is the Base area of collecting of tank.

'h'= the height of water collected, in mm = 100 mm

't_w' = time in seconds for collecting 100 mm high water in the collecting tank.

3. Power in put to the pump, IP

$$IP \text{ KW} = \frac{20}{1200} \ x \ \frac{60 \ x \ 60}{T} \ x \ 0.75$$

Where: 'T' is time in seconds for 20 revolutions of energy meter disc.
0.75 is the mechanical efficiency of the pump and motor.
1200 is the energy meter constant.

4. Output Power (Delivered by the pump), BP

$$BP = \frac{\rho \, g \, Q \, H}{1000} \text{ KW.}$$

Where, 'ρ' = 1000 Kg/ m^3

'Q' = from Formula 2

'H' = from Formula 1

'g' = 9.81 m/s^2.

The results for Main Characteristics are tabulated as shown in Tabular Column – 2.

Efficiency, $\eta = \frac{BP}{IP}$

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Tabular Column – 1, Test Readings for Main Characteristics of Centrifugal Pump

Reading set	Speed, rpm	Trial No:	Suction Head, H_v, mm	Delivery head, H_d, Kg/cm^2	Time, 't_w 'in seconds for filling 100 mm height of water in collecting tank	Time, 'T' in seconds for 20 revolutions of energy meter disc.
A	2275	1				
		2				
		3				
		4				
		5				
B	2475	1				
		2				
		3				
		4				
		5				
C	2575	1				
		2				
		3				
		4				
		5				

Tabular Column – 2, Results for Main Characteristics of Centrifugal Pump

Reading set	Speed, rpm	Trial No:	Discharge, Q, m^3/s	Total Head, H, in meters of water	Power Input to pump, IP, kW	Power Output of Pump, BP, kW	Efficiency, η, %
A	2275	1					
		2					
		3					
		4					
		5					
B	2475	1					
		2					
		3					
		4					
		5					
C	2575	1					
		2					
		3					
		4					

		5					

Ex.No.12 Date:	PERFORMANCE TESTS ON A PELTON WHEEL TURBINE

Aim: To obtain main and operating characteristics of Pelton Turbine.

Specifications of Pelton Turbine:

1. High head water supply pump motor capacity : 7.5 hp, 3 ph, 440 V, 50 Hz, ac

2. Details of Turbine-

Mean diameter	: 250 mm
No. of buckets	: 20
Water jet diameter	: 18 mm
Turbine runaway speed	: 2000 rpm
Maximum flow rate	: 250 liters per minute
Maximum head	: 75 m
Maximum shaft output	: 1.5 hp

3. Details of Loading-

Brake drum
Brake drum radius + rope diameter, R = 0.135 m

4. Details of discharge measurement set up-

Type	: V- notch
Angle of V- notch	: 90°
Coefficient of discharge	: 0.6

Operating Procedure:

The centrifugal pump supplies water at high head to Pelton Turbine. The discharge of the centrifugal pump is controlled with the delivery valve. The quantity of water supplied to the turbine is controlled using the spear valve. Turbine discharge is measured using V-notch. Loading of turbine is carried out using brake drum

C. To Obtain Main Characteristics

This test is carried out by running the turbine at different loads. The rpm of the turbine is not maintained constant.

Step – 1: Note down the V-notch upstream water level. Use this data to calculate head over notch.

Step – 2: Keep the centrifugal pump delivery gate valve closed.

Step – 3: Keep the spear valve at minimum position.

Step – 4: Release the ropes over the brake drum by operating the hand wheel so that the turbine is on no load.

Step – 5: Switch on the centrifugal pump and open the gate valve.

Step – 6: Adjust the spear valve so that the speed of the turbine is approximately 1950 rpm.

Step – 7: Note down the water level at V-notch upstream, turbine rpm, readings of brake drum spring balances.

Step – 8: Adjust the hand wheel of the brake drum spring balance so that the spring balance reads 1.0 kg. Speed of turbine will reduce. Note down pressure gauge reading at turbine inlet, the water level at V-notch upstream, turbine rpm, readings of brake drum spring balances.

Step – 9: Repeat Step – 8 for spring balance readings 2.0 kg, 2.5 kg, 3.0 kg, 3.5 kg, 4.0 kg, 4.5 kg and 5.0 kg. Note down the corresponding values of pressure gauge reading at turbine inlet, the water level at V-notch upstream, turbine rpm, readings of brake drum spring balances.

Tabulate the readings as shown in Tabular Column – 1.

D. To Obtain Operating Characteristics

This test is carried out by running the turbine at constant speed at different loads.

Step – 1: Note down the V-notch upstream water level before the start of experiment. Use this data to calculate head over notch.

Step – 2: Keep the centrifugal pump delivery gate valve closed.

Step – 3: Keep the spear valve at minimum position.

Step – 4: Release the ropes over the brake drum by operating the hand wheel so that the turbine is on no load.

Step – 5: Switch on the centrifugal pump.

Step – 6: Open fully the centrifugal pump delivery gate valve.

Step – 7: Operate the spear valve so that the turbine speed is 1250 rpm.

Step – 8: Note down the head on turbine, water level at V-notch upstream, turbine rpm, readings of brake drum spring balances.

Step – 9: Adjust the hand wheel of the brake drum spring balance so that the spring balance reads 1.5 kg. Speed of turbine will reduce. Adjust the spear valve so that the speed of the turbine is brought back to 1250 rpm. Note down pressure gauge reading at turbine inlet, water level at V-notch upstream, turbine rpm, readings of brake drum spring balances.

Step – 10: Repeat Step – 7 for loads 3.0 kg, 4.5 kg and 6 kg. Adjust the spear valve so that the speed of the turbine is brought back to 1250 rpm. Note down pressure gauge reading at turbine inlet, water level at V-notch upstream, turbine rpm, readings of brake drum spring balances.

Step – 11: Release the load on the brake drum by operating he hand wheel. While releasing the load, adjust the spear valve so that speed of the turbine is not exceeding 2000 rpm.

Step – 12: At no load condition of turbine, adjust the spear valve so that the turbine speed is 1500 rpm. Repeat Steps 6, 7, 8 and 9.

Step – 13: Release the load on the brake drum by operating he hand wheel. While releasing the load, adjust the spear valve so that speed of the turbine is not exceeding 2000 rpm.

Step – 14: At no load condition of turbine, adjust the spear valve so that the turbine speed is 1750 rpm. Repeat Steps 6, 7, 8 and 9.

Tabulate the readings as given in Tabular Column - 3

FORMULAE FOR CALCULATIONS

1. Net head on Turbine
   'H' in meters of water = 10 x P

   Where 'P' is the pressure

   Gauge reading in Kg/ cm^2

2. Head over notch, 'h' m = $(H_2 - H_1)/1000$

3. Discharge (Flow Rate)
   Water through the Turbine = Flow Rate by V-Notch

$$Q = \frac{8}{15} C_d \sqrt{2g} \tan\frac{\theta}{2} h^{\frac{5}{2}} \quad m^3/s$$

u

Where,

Coefficient of discharge, C_d = 0.6

'h' = Head over the notch in **meters.**

θ = 90^0.

g = 9.81 m/s^2

4. Hydraulic Power Input to the Turbine, $HP_{hyd} = \rho \times Q \times H \times 9.81$ Watts

Where, ρ, density of water = 1000 Kg/m^3

Q, discharge = Flow rate of water in m3 /sec. from Formulae-2.

H, Head over turbine = Head on Turbine in meters from Formulae-1.

4. Brake Horse Power of the Turbine, BHP

Torque acting on the brake drum, $T = (F_1-F_2) \times R \times 9.81$ N-m

Where F_1 and F_2 are the spring balance reading and R is the sum of brake drum radius and rope diameter.

R = 0.135 m (from specification)

Brake horse power of turbine, BHP = $\frac{2\pi N T}{60}$ *Watts*

5. Turbine Efficiency, % η_{tur} = BHP / HP_{hyd} x 100

6. Unit Quantities – Under Unit Head,

a) Unit Speed, N_u = $N/\sqrt{H}$

b) Unit Power, P_u = $P / H^{2/3}$

c) Unit Discharge, Q_u = $Q/\sqrt{H}$

7. Specific Speed

$$N_s = \frac{N\sqrt{P}}{H^{5/4}}$$

For calculating specific speed, power to be taken in kW.

Tabular Column – 1, Test Readings for Main Characteristics of Pelton Turbine

V-notch upstream water level before the start of experiment, **H_1**= mm

Trial No:	Speed, rpm	Brake drum reading		V-notch upstream water level, H_2, mm	Head on turbine, P, kg/cm^2
		F_1, kg	F_2, kg		
1		0	0		
2		1.0			
3		2.0			
4		3.0			
5		4.0			
6		5.0			

Tabular Column – 2, Results for Main Characteristics of Pelton Turbine

Trial No:	Speed, rpm	Net head on turbine H, m	Discharge, Q, m^3/s	Power Input ot Turbine, HP $_{hyd}$	Brake horse power of turbine, BHP	Turbine Efficiency, $\eta = \frac{BHP}{HP_{hyd}} x100$
1					0	0
2						
3						
4						
5						
6						

Tabular Column – 3, Test Readings for Operating Characteristics of Pelton Turbine

V-notch upstream water level before the start of experiment, $\mathbf{H_1}$= mm

Speed, rpm	Trial No:	Brake drum reading		V-notch upstream water level, H_2, mm	Head on turbine, P, kg/cm^2
		F_1, kg	F_2, kg		
1250	1	0	0		
	2	1.5			
	3	3.0			
	4	5.0			
	5	6.0			
1500	6	0	0		
	7	1.5			
	8	3.0			
	9	5.0			
	10	6.0			
1750	11	0	0		
	12	1.5			
	13	3.0			
	14	5.0			
	15	6.0			

Tabular Column – 4, Results for Operating Characteristics of Pelton Turbine

Trial No:	Speed, rpm	Net head on turbine H, m	Discharge, Q, m^3/s	Power Input to Turbine, HP_{hyd}	Brake Horse Power of Turbine, BHP	Turbine Efficiency, $\eta = \frac{BHP}{HP_{hyd}} x100$
1	1250				0	0
2						
3						
4						
5						
6	1500					
7						
8						
9						
10						
11	1750					
12						
13						
14						
15						

Result:The main and operating characteristics of Pelton Wheel has been obtained.

ROUGH WORK

Ex.No.13 Date:	PERFORMANCE TESTS ON FRANCIS TURBINE

Aim: To determine the operating and main characteristics of Francis Turbine

SPECIFICATIONS:

* ***Turbine*** : 150mm Dia Impeller.

Maximum Flow of water – 1200 lpm (approx.)

Maximum Head – 20 mts (approx)

* ***Loading*** : AC Alternator.

* ***Provisions***

a) Flow rate measurement using Venturimeter,

b) Supply water control by Butterfly Valve.

A. TO OBTAIN CONSTANT SPEED CHARACTERISTICS:

(Operating Characteristics)

1. Keep the gate opening partially, and set the guide vane opening so that turbine speed is 1300 rpm.

2. For different electrical loads on the turbine / generator, change the gate opening, so that the speed is held constant. See that the voltage does not exceed 250 V to avoid excess voltage on bulbs.

3. Change the guide vane opening settings to different positions and repeat (2) for different speeds 1200 rpm, 1100rpm and tabulate the results.

Tabulate the readings in Table -1

B. TO OBTAIN CONSTANT HEAD CHARACTERISTICS:

(Main Characteristics)

1. Select the guide vane angle position.
2. Keep the gate closed, and start the pump.
3. Slowly open the gate and set the pressure on the gauge.
4. For different electrical loads, change the gate valve position, and maintain the constant head and tabulate the results as given in Table – II.

FORMULAE FOR CALCULATIONS

DATA:

Venturimeter

Inlet Dia, 'D' = 100mm

Throat Dia, 'd' = 60mm

Cd = 0.9

Density of Water = 1000 Kg/m^3

Energy Meter Constant = Rev. / KWH

1.Head on the Turbine 'H' in m of water = 10 (P +P_v / 760)

Where 'P' is the pressure gauge reading in Kg/cm^2 and 'P_V' is the Vacuum Gauge reading.

2. Discharge (Flow Rate) of water through the Turbine = flow rate through Venturimeter .

$$Q = C_d \frac{a_1 a_2 \sqrt{2gh}}{\sqrt{a_1^2 a_2^2}} m^3/s$$

Where, C_d = 0.9

a_1 =Area of Inlet Section of Venturi = $\pi D^2 / 4$ = π x $(100/1000)^2/4$ m^2.

a_2 =Area of Out Section of Venturi (Throat) = $\pi d^2 / 4$ = π x $(60/1000)^2/4$ m^2.

Loss of Head, h = (Pi – Pt) x 10.0 mtrs (because pressure of 1 kg/cm^2 = head of 10.0 m of water head)

Where, $(P_i - P_t)$ = Differential Head across Venturimeter in Kg/cm^2.

1. Input power (water power) of Turbine '**IP_{tur}**'

$$= \frac{\rho g Q H}{1000} \quad \textbf{KW}$$

Where ρ =1000 Kg /m^3

Q =Discharge flow rate in m^3 /Sec.

H Net head on turbine in meters.

4. out power to turbine 'BP'_{tur} =

$$\frac{5}{energy\ meter\ cons\tan t} x \frac{60\,x\,60}{t} x \frac{1}{0.75} \text{KW}$$

t= Time taken in sec. for 5 revolutions of energy meter disk

0.75 = Efficiency of the alternator

5. Turbine efficiency $\%\eta_{tur}$ = $\frac{BP_{tur}}{IP_{elec.}}$

Result:

The operating and main characteristics of Francis Turbine is obtained.

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TABLE OF READINGS- 1

Constant Speed Characteristics.**(Operating Characteristics)**

METHOD : **By Keeping the guide vane opening constant &By changing the Gate Valve position.**

Turbine speed in RPM	Head on Turbine		Venturimeter		Diff. head across Venturimeter (P_i-P_t) in Kg/cm^2	Load on Generator		Wattage of Bulb in action, W	Energy Meter Reading Time for 5 Revlns. In seconds.
	Pressure "P" in Kg/cm^2	Vacuum "P_v" in mm of Hg.	Inet Pressure " P_i" in Kg/cm^2	Throat Pressure "P_t"in Kg/cm^2		"V" in Volts	" I " in Amps		
1300									
1200									
900									

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TABLE OF READINGS- II

Constant Head Characteristics.

METHOD : **By Keeping the guide vane angle constant & by changing the Gate Valve position.**

Guide Vane Valve Position	Turbine speed in RPM	Head on Turbine		Venturimeter		Diff. head across Venturimeter (P_i-P_t) in Kg/cm^2	Load on Generator		Wattage of Bulb in action, W	Energy Meter Reading Time for 5 Revlns. In seconds.
		Pressure "P" in Kg/cm^2	Vacuum "P_v" in mm of Hg.	Inet Pressure " P_i" in Kg/cm^2	Throat Pressure "P_t"in Kg/cm^2		"V" in Volts	" I " in Amps		
Position 1	1500								0	
Position 2	1400									
Position 3										

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TABLE OF CALCULATIONS -1

Constant Speed Characteristics

Turbine Speed in RPM	**Net Head on Turbine 'H' in mtrs.**	**Discharge (Flow Rate) 'Q' in m^3 /Sec.**	**IP_{elec} In KW**	**BP_{tur} In KW**	**% η_{tur}**

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TABLE OF CALCULATIONS -II

Constant Head Characteristics

Guide Vane Valve Position	**Turbine Speed in RPM**	**Net Head on Turbine 'H' in mtrs.**	**Discharge (Flow Rate) 'Q' in m^3 /Sec.**	**IP_{elec} In KW**	**BP_{tur} In KW**	**% η_{tur}**
Position -1						
Position 2						
Position 3						

Ex.No.14 Date:	PERFORMANCE TESTS ON KAPLAN TURBINE

Aim: To study the performance of Kaplan Turbine and to draw Main Characteristics at constant speed and Operating Characteristics at constant head.

Turbine Specifications:

1. Maximum rater power – 4 kW.
2. Impeller diameter - 150 mm
3. Maximum allowable speed (run away speed) – 3000 rpm.
4. Discharge measurement – Rectangular notch. Coefficient of discharge – 0.6, Breadth 'b' of rectangular notch – 250 mm.
5. Loading of turbine – Electrical loading. Energy meter constant- revolutions per kWhr.

General Operating Procedure:

Step – 1: Keep the centrifugal pump delivery gate valve closed. Keep the turbine rotor blade opening valve to minimum position by operating the hand wheel.

Step – 2: Switch on the centrifugal pump.

Step – 3: Set the centrifugal pump delivery gate valve to maximum opening position.

Step – 4: By operating the hand wheel, open the rotor blade opening valve slightly. The turbine will start to rotate. Note the voltage meter on the alternator output. Ensure that the voltage does not exceed 250 V by limiting the rpm of the turbine.

Step – 5: Depending on the load, rotor blade opening valve can be adjusted to achieve required speed of rotation of the turbine.

A. To Obtain Constant Speed Characteristics (Operating Characteristics).

Follow steps 1 -4 of general operating procedure.

Step – 5: Ensure that the turbine is on "No-Load". Open rotor blade opening valve so that the turbine rpm is 1500. If the voltage developed by the alternator at this rpm is more than 250 V, select a lower rpm so that the voltage developed is less that 250 V. Note down the turbine speed in rpm, Inlet pressure 'P', Outlet pressure 'Pv', Time 't' for 4

revolutions of energy meter in seconds, Head over notch 'h' in mm, Voltmeter reading, Ammeter reading and Number of engaged switches in alternator output.

Step – 6: Engage 2 switches in the alternator output. The speed of rotation of turbine will reduce. By operating the hand wheel increase the opening of the rotor blade opening valve so that the rpm of the turbine is1500. After waiting 5 (approx.) minutes for flow stablisation, note down the reading as explained in step – 5.

Step – 7:

Engage 2 more switches and after matching the turbine rpm and after flow stablisation, note down the readings.

Step – 8: Continue Step – 7 till all the switches are engaged.

Operate the hand wheel and close the rotor blade opening valve. Disengage all switches on alternator output and ensure that turbine is on "No Load" condition.

Step – 9: Open the rotor blade opening valve by operating the hand wheel so that the turbine speed is 1300 rpm. Repeat the steps explained and note down the readings and ensure that the turbine is on "No Load" condition after completing the test.

Step – 10: Open the rotor blade opening valve by operating the hand wheel so that the turbine speed is 1000 rpm. Repeat the steps explained and note down the readings and ensure that the turbine is on "No Load" condition after completing the test.

Tabulate the readings as given in the following tables.

Constant Speed Characteristics (Operating Characteristics).

Turbine speed in RPM	No. of engaged switches	Head on Turbine		Energy meter reading				
		Pressure "P" in Kg/cm^2	Vacuum "P_v" in mm of Hg.	no. revolution of disk "n"	Time taken in sec. "t"	Voltmeter reading 'V' volts	Ammeter Reading 'I' amps	Head over notch 'h' in mm
1500	0			4				
	2			4				
	4			4				
	6			4				
	8			4				
	10			4				
1300	0			4				
	2			4				
	4			4				
	6			4				
	8			4				
	10			4				

1000	0			4				
	2			4				
	4			4				
	6			4				
	8			4				
	10			4				

TABLE OF CALCULATIONS -1 Constant Speed Characteristics

Turbine Speed in RPM	Net Head on Turbine 'H' in mtrs.	Discharge (Flow Rate) 'Q' in m^3 /Sec.	IP_{elec} In KW	BP_{tur} In KW	% η_{tur}	% of Full Load
1500						
1300						

1000						

B. To Obtain Constant Head Characteristics (Main Characteristics)

Follow steps 1 and 2 of general operation procedure.

Step – 3: Set the centrifugal pump delivery gate valve to ¼ opening. Open the rotor blade opening valve by operating the hand wheel. The turbine speed increases and the voltage developed by the alternator also increases. Set the valve position in such a way that the voltage developed by the alternator is approximately 250 V. Note down the rpm of the turbine. Let it be N1.

Note down the turbine speed in rpm, Inlet pressure 'P', Outlet pressure 'Pv', Time 't' for 4 revolutions of energy meter in seconds, Head over notch 'h' in mm, Voltmeter reading, Ammeter reading and Number of engaged switches inalternator output.

Step – 4: Engage 2 switches in the alternator output so that turbine is loaded and the speed of turbine will decrease. Note down the readings as explained in Step – 3. If there is a drop in turbine inlet pressure, adjust the gate valve so that the pressure is maintained.

Step – 5: Engage 2 more switches and note down the readings.

Step – 6: Continue Step – 5 until all the switches are engaged and readings are taken.

Close the rotor blade opening valve. Disengage all switches on the alternator output.

Step – 7: Set the centrifugal delivery valve at ½ position. Open the rotor blade opening valve so that the turbine rpm is 150 less than N1. Let the new rpm be N2. (N1 is explained in Step – 3). Repeat Step – 4 and Step – 5.

Step – 8: Set the centrifugal pump delivery valve at ¾ position. Open the rotor blade opening valve so that the turbine rpm is 150 less than N2. Let the new rpm be N3. Repeat Step – 4 and Step – 5.

Step – 9: Set the centrifugal pump delivery valve at full opening. Open the rotor blade opening valve so that the turbine rpm is 150 less than N3. Repeat Step – 4 and Step – 5.

Constant Head Characteristics (Main Characteristics).

Head on Turbine			No. of engaged switches	Energy meter reading				
Pressure "P" in Kg/cm^2	Vacuum "P_v" in mm of Hg.	Turbine speed in RPM		no. revolution of disk "n"	Time taken in sec. "t"	Voltmeter reading 'V' volts	Ammeter Reading 'I' amps	Head over notch 'h' in mm
			0	4				
			2	4				
			4	4				
			6	4				
			8	4				
			10	4				
			0	4				
			2	4				
			4	4				
			6	4				
			8	4				
			10	4				

			0	4				
			2	4				
			4	4				
			6	4				
			8	4				
			10	4				
			0	4				
			2	4				
			4	4				
			6	4				
			8	4				
			10	4				

TABLE OF CALCULATIONS -II Constant Head Characteristics

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Net Head on Turbine 'H' in mtrs.	Turbine Speed in RPM	Discharge (Flow Rate) 'Q' in m^3 /Sec.	IP_{elec} In KW	BP_{tur} In KW	% η_{tur}	% of Full Load

TABLE OF CALCULATIONS -II Constant Head Characteristics

Net Head on Turbine 'H' in mtrs.	Turbine Speed in RPM	Discharge (Flow Rate) 'Q' in m^3 /Sec.	IP_{elec} In KW	BP_{tur} In KW	% η_{tur}	% of Full Load

FORMULAE FOR CALCULATIONS

1. Out power to turbine 'BP'_{tur} = $\frac{n}{energy\ meter\ cons\tan t} x \frac{60\,x\,60}{t} x \frac{1}{0.75}$ KW

where n = No. of revolutions of energy meter disk

t = Time taken in sec. of energy meter disk

0.75 = Efficiency of the alternator is 75%

2.Head on the Turbine 'H' in m of water = 10 ($P + P_v / 760$)

Where 'P' is the pressure gauge reading in Kg_f/cm^2 and 'P_V' is the Vacuum Gauge reading in mm of Hg.

3. Discharge (Flow Rate) of water through the Turbine = flow rate of water through notch.

Q = 2/3 C_d b $\sqrt{2g}$ $h^{3/2}$ m^3 /s.

Where, b = breadth of notch = 250mm=0.25m

h = head over the notch in m

C_d = Co-efficient of discharge = 0.6

4. Input power (water power) to Turbine 'IP$_{tur}$'

$$= \frac{\rho g Q H}{1000} \text{ kW}$$

Where ρ = 1000 Kg /m^3

Q = Discharge flow rate in m^3 /s.

H = Net head on turbine in meters

g = =9.81

5. Turbine efficiency $\%\eta_{tur}$ = $\frac{BP_{tur}}{IP_{elec.}}$

6. Unit Quantities – Under Unit Head,

d) Unit Speed, N_u = N/ $\sqrt{H}$

e) Unit Power, P_u = P / $H^{2/3}$

f) Unit Discharge, Q_u = Q / $\sqrt{H}$

7. Specific Speed

$$N_s = \frac{N\sqrt{P}}{H^{5/4}}$$

Obtained at maximum efficiency.

Where N, H, P are picked out from the table of the calculation

Result:

The main and operating characteristics of Kaplan Turbine has been obtained.

Rough Work